

Motor Datasheet: WxF-85x22SL

Machine Configuration

Parameter	Value	Unit
Machine Outer Diameter	83.0	mm
Machine Length / Thickness	21.7	mm
Voltage	48.0	V
Operating Temperature	-40 to 85	°C

Mass Properties

Parameter	Value	Unit
Machine Mass (active components only)	492.5	g
Rotor Mass	169.3	g
Stator Mass	323.1	g
Rotor Structure Included	No	-
Rotor Moment of Inertia	1763.4	g·cm ²

Performance Metrics

Parameter	Value	Unit
Torque (continuous)	3.92	Nm
Speed	2500	rpm
Losses (continuous)	83.5	W
Supply Current (RMS)	18.79	A
Phase Resistance (per phase)	55.7	mΩ
Terminal Inductance ($L_{tt} = L_d + L_q$)	250.2	μH
D-axis Inductance (L_d)	126.5	μH
K_e (peak, phase)	9.54	mV/rpm
K_e (peak, line-to-line)	16.52	mV/rpm
K_t (peak)	136.7	mNm/A
Motor Constant (K_m)	511.0	mNm/√W